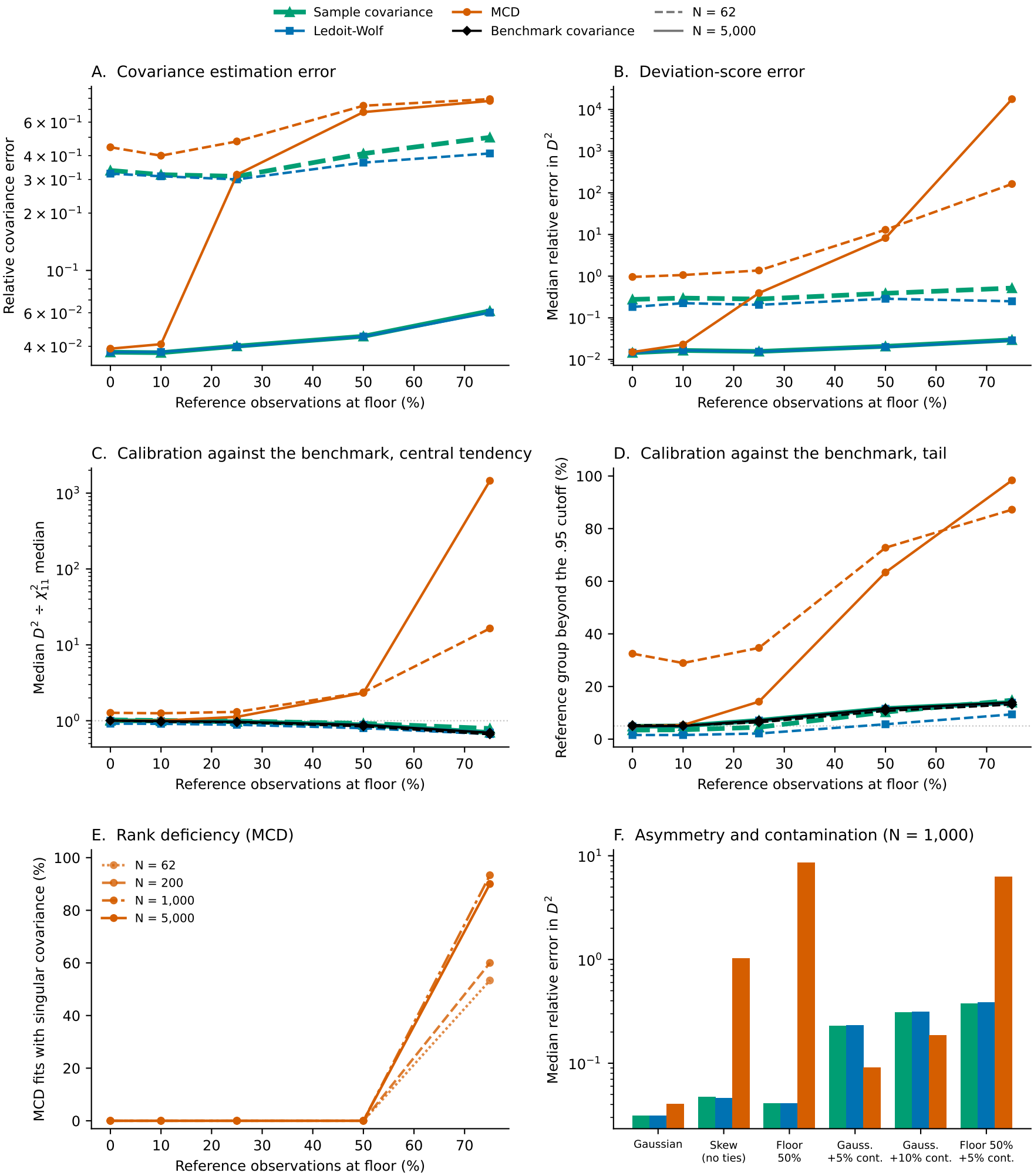
Sample covariance
Ledoit-Wolf
MCD
Benchmark covariance
N = 62
N = 5,000
A. Covariance estimation error
Relative covariance error
Reference observations at floor (%)
B. Deviation-score error
Median relative error in D^2
Reference observations at floor (%)
C. Calibration against the benchmark, central tendency
Median $D^2 \div \chi^2_{11}$ median
Reference observations at floor (%)
D. Calibration against the benchmark, tail
Reference group beyond the .95 cutoff (%)
Reference observations at floor (%)
E. Rank deficiency (MCD)
N = 62
N = 200
N = 1,000
N = 5,000
MCD fits with singular covariance (%)
Reference observations at floor (%)
F. Asymmetry and contamination (N = 1,000)
Median relative error in D^2
Gaussian
Skew (no ties)
Floor 50%
Gauss. +5% cont.
Gauss. +10% cont.
Floor 50% +5% cont.